Renderoc LA

construction solutions

Single component free flowing low alkali micro concrete

Uses

The high fluid nature of Renderoc LA obviates the need for compaction and vibration and are most suitable for:

- Repair zones with restricted access or congested reinforcement
- Reinstatement of large, strctural sections of concrete
- Repairs to concrete effected by alkali-silica reaction
- Areas where excellent chloride and carbon-di-oxide resistance is required

Advantages

- Dual expansion system compensates for shrinkage in the plastic and hardened states.
- Low alkali content minimises risk of alkali-silica reaction
- Exceptional bond to concrete substrates without independent primer
- Suitable for placement by pumping or pouring techniques into restricted locations
- Self-compacting nature eliminates honeycombing and displaces air without vibration
- High strength and low permeability provide maximum protection against carbon-di-oxide and chlorides
- Pre-bagged to overcome site-batched variations - only the site addition of clean water is required
- Contains no chloride admixtures

Description

Renderoc LA is supplied as a ready to use blend of dry powders which requires only the site addition of clean water to produce a free-flowing, shrinkage compensated micro-concrete suitable for large volume concrete repairs at nominal thicknesses in excess of 50mm.

Specification

The fluid repair system (micro-concrete) shall be Renderoc LA, a single component, low-alkali cement-based blend of powders to which only the site-addition of clean water shall be permitted. The micro-concrete shall exhibit compressive strength of not less than 45N/mm² at 7 days and not less than 60N/mm² at 28 days and Flexural strength of not less than 7N/mm² at 28 days.

Properties

The following typical results were obtained at a water to powder ratio of 0.135 and temperature of 20ºC.

Test method		Typical result
Compressive strength (BS 1881 Pt.116[1])	:	> 28 N/mm² at 3 days > 45 N/mm² at 7 days > 60 N/mm² at 28 days
Flexural strength (BS 1881: Pt 117)	:	> 7 N/mm² at 28 days
Bond strength, 28 days ASTM C1583 (50mm dolly)	:	>1.0 MPa
Rapid Chloride Penetration , 28 days ASTM C1202	:	1100 coulombs (low) pass
Water permeability DIN1048 Part 5	:	< 10 mm
Note	:	The low permeability of Renderoc LA retards chemical attack in aggressive environments. The cured micro-concrete is resistant to acid gases, chloride ions, oxygen and water. For specific data regarding chemical resistance, contact the local Fosroc office.

Design criteria

Renderoc LA is designed for large volume repairs typically in excess of 50mm deep. The product can be applied at sections generally up to 150 mm thick although greater thicknesses may be achievable dependent on the repair geometry. Attention to the basic design criteria given below should ensure that the full benefits of this technology are gained in use:

(i) Water addition = 2.8 - 3.2 ltrs / 25kg bag under no circumstances should part bags be used or additional water be employed. Either of these two actions will adversely affect material performance, automatically invalidating Fosroc's standard product guarantee.

Method Statement

Refer to your local Fosroc office for the method statement specifying surface preparation, priming, mixing, application and cleaning methods.

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Estimating

Supply

Renderoc LA	:	25kg bags

Coverage

Actual coverage rate is approximate, and will be wholly dependent upon the general substrate condition. It is recommended, therefore, that site trials over a typical area are conducted to determine the likely usage/coverage rates for a particular substrate.

Renderoc LA	:	12 litres/25kg bag

Limitations

- Renderoc LA should not be used when the ambient temperature is below 5˚C and falling.
- Renderoc LA should not be part mixed.
- Renderoc LA should not be exposed to running water either during application or prior to final set.
- Water ponding is not recommended for curing.

Storage

Shelf life

Renderoc LA has a shelf life of 12 months; if kept in a dry environment, in its original, unopened packing. If stored in conditions of high humidity and/or temperature, the shelf life of Renderoc LA will be reduced to as little as 4 months.

Precautions

Health and safety

Renderoc LA contains cement powders which, when mixed with water or upon becoming damp, release alkalis which can be harmful to the skin.

During use, avoid inhalation of the dust and contact with the skin or eyes. Wear suitable protective clothing - eye protection, gloves and respiratory equipment (particularly in confined spaces).

The use of barrier creams to provide additional skin protection is also advised. In case of contact with the skin, rinse with plenty of clean water, then cleanse thoroughly with soap and water.

In case of contact with eyes, rinse immediately with plenty of clean water and seek medical advice. If swallowed seek medical attention immediately - do not induce vomiting.

Fire

Renderoc LA is non-flammable and thus presents no fire hazard.

For further information, please refer to the Product Material Safety Data Sheets.

Important note

Fosroc products are guaranteed against defective materials and manufacture and are sold subject to its standard Conditions for the Supply of Goods and Service. **All Fosroc datasheets are updated on a regular basis. It is the user's responsibility to obtain the recent version.**

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